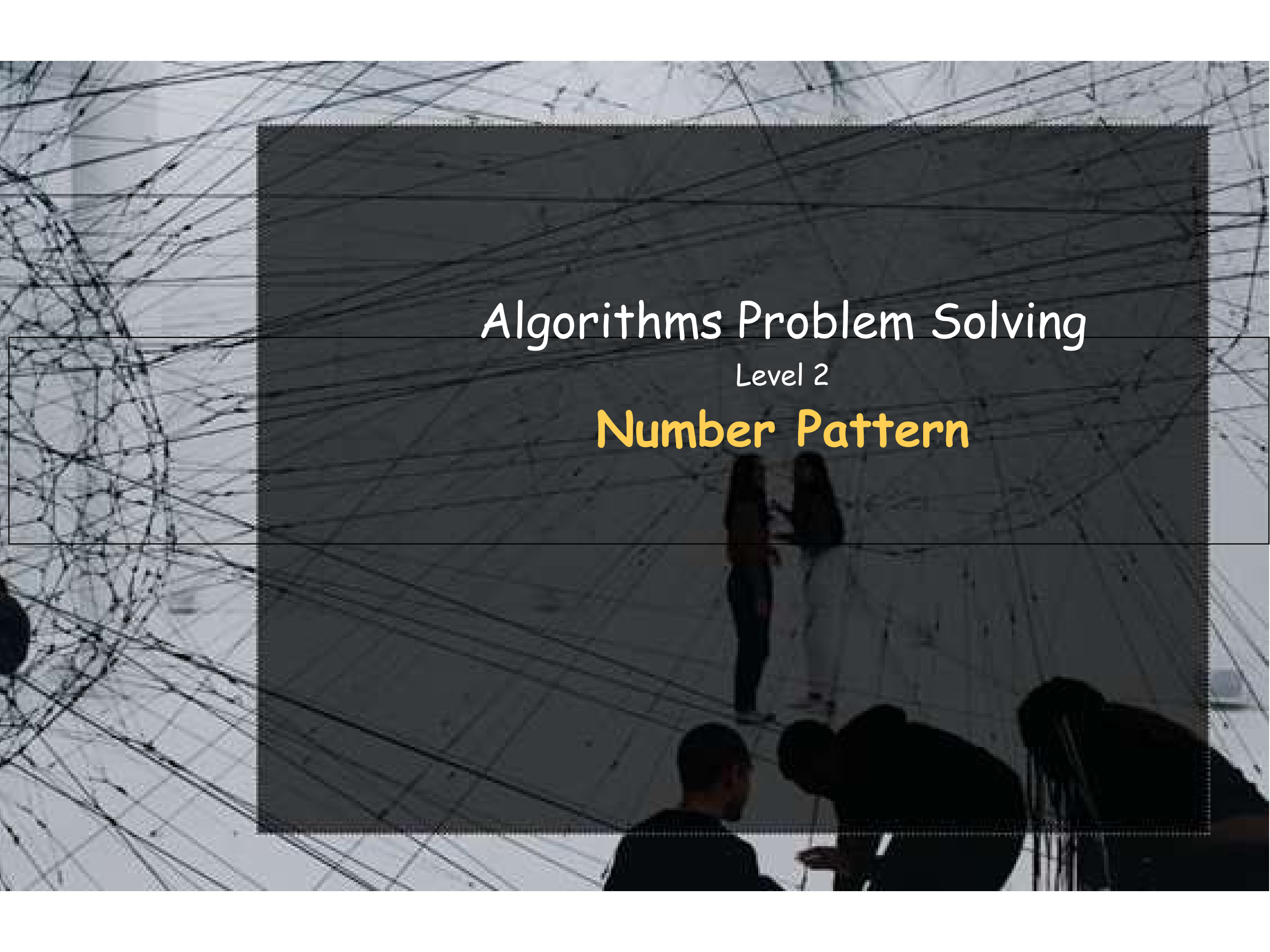


Algorithms Problem Solving

Level 2

Number Pattern



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Definition

- ◆ A number pattern is a pattern where each row prints the same number multiple times.
- ◆ In this program, the numbers increase from `1` to the entered number.
- ◆ The number of repetitions also increases with each row.
- ◆ For example, input `3` produces `1`, `22`, `333`.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``PrintInvertedPattern()`` uses two nested ``for`` loops.
- ◆ The outer loop controls the current number and the number of rows.
- ◆ It starts from ``1`` and continues until ``Number``.
- ◆ The inner loop prints the current value of ``i``.
- ◆ The inner loop runs ``i`` times.
- ◆ After the inner loop finishes, ``endl`` moves to the next line.
- ◆ As ``i`` increases, both the number being printed and the number of repetitions increase.

Steps in main()

1. Read a non-negative number.
2. Start the outer loop from `1` to `Number`.
3. Start the inner loop from `1`.
4. Print `i` while $j \leq i$.
5. Move to the next line.
6. Increase `i` and repeat.

💡 Example

◆ Input:

`5`

◆ First row: `i = 1` → print `1` one time.

◆ Second row: `i = 2` → print `2` two times.

◆ Third row: `i = 3` → print `3` three times.

◆ Fourth row: `i = 4` → print `4` four times.

◆ Fifth row: `i = 5` → print `5` five times.

➡ Result

1

22

333

4444

55555

Note

- ◆ The outer loop controls the rows and determines which number is printed.
- ◆ The inner loop controls how many times the current number is printed.
- ◆ For every row, the value of `i` is used for both the number printed and the number of repetitions.

⊘ Warning

- ◆ The validation allows `0` because the condition is $N < 0$.
- ◆ If the user enters `0`, the outer loop does not run and nothing is printed.



Comparison

▨ #12_Inverted_Number_Pattern

- ✓ Starts from `Number` and counts down to `1`.
- ✓ Example: `5 → 55555, 4444, 333, 22, 1`.

▤ #13_Number_Pattern

- ✓ Starts from `1` and counts up to `Number`.
- ✓ Example: `5 → 1, 22, 333, 4444, 55555`.

✦✦ Summary

- ✓ Use an outer loop from `1` to `Number`.
- ✓ Use an inner loop to repeat the current number.
- ✓ The value of `i` determines both the digit and the repetition count.
- ✓ Each row contains one more digit than the previous row.
- ✓ Input `5` produces `1`, `22`, `333`, `4444`, `55555`.

Thank you 😊